

Milestone 5 Follow-up Analyses

Responses to InComm technical review questions — Model A (InComm-sourced)

Purpose

This report answers the follow-up questions from the Milestone 5 review. Where training was required, Model A was retrained across multiple random seeds (42, 123, 456, 789, 2026) using a member-grouped split. Headline v1 metrics in the original five deliverables used seed 42; this document shows whether those results are stable.

1. Closure rate and class imbalance

Metric	Value
Model A rows	11,669
Unique members	6,668
Closed / Open	4,726 / 6,943
Closure rate	40.5%
Imbalance	moderate (not rare-event)
Handling	class_weight='balanced' in the logistic pipeline; no downsampling or synthetic labels

2. Column contract and missingness

The analytical dataset used for Milestone 5 has 28 columns (loader requires 28). The fitted model uses 20 features after dropping identifiers, the label, and leaking qualifying-claim fields. There is no 44-column contract in this codebase.

Metric	Value
Analytical dataset columns	28
Loader required columns	28
Model features	20

Rows are not dropped. No sentinel values. Numeric missings are median-imputed inside the training pipeline (SimpleImputer).

Feature	N Rows Missing	Pct Rows Missing	Pct Members With Any Missing
days_since_gap_opened	0	0.0%	0.0%
prior_closure_rate_others_measures	1042	8.9%	15.6%

rewards_issued_count	0	0.0%	0.0%
rewards_claimed_count	0	0.0%	0.0%
redemption_rate	0	0.0%	0.0%
avg_days_issue_to_claim	5353	45.9%	55.4%

Sensitivity: retraining after dropping rows missing prior_closure_rate_other_measures (dropped 1,042 rows) yields mean AUC 0.9273 ± 0.0073 versus impute-all mean AUC 0.9411 ± 0.0030 .

3. Reward features at prediction time (cold start)

Never-rewarded members receive rewards_issued_count = 0, rewards_claimed_count = 0, and redemption_rate = 0. avg_days_issue_to_claim is undefined until a claim occurs and is median-imputed. The table below is test-set performance on the last baseline seed.

Group	N Test	Closure Rate	Mean Predicted Probability	AUC
never_rewarded	425	0.0%	0.0611	—
has_reward_history	1953	48.6%	0.5177	0.9176

Ablation (same seeds, reward features removed): mean AUC 0.7477 ± 0.0077 . Drop vs full feature set: 0.1934. A large drop means the 0.94-range result is engagement-driven and still depends on reward aggregates being computed as of scoring time, not after gap closure.

4. Grouped split vs care-gap stratification

The production split groups on member_id so a member cannot appear in both train and test. It is not stratified by care_gap_code by default. The tables show mix for the last seed, then mean AUC across seeds for grouped vs grouped+stratified.

Grouped only

Care Gap Code	N Train	N Test	Pct In Test
BH_POST_DISCHARGE_7D	1317	350	21.0%
COPD_SPIROMETRY	1337	330	19.8%
DIAB_A1C_TEST	1319	348	20.9%
DIAB_BLOOD_SUGAR	1319	348	20.9%
DIAB_EYE_EXAM	1319	348	20.9%
DIAB_NEPHRO_SCREEN	1319	348	20.9%
HTN_PCP_FOLLOWUP	1361	306	18.4%

Grouped + stratified by care gap

Care Gap Code	N Train	N Test	Pct In Test	
BH_POST_DISCHARGE_7D	1334	333	20.0%	
COPD_SPIROMETRY	1334	333	20.0%	
DIAB_A1C_TEST	1333	334	20.0%	
DIAB_BLOOD_SUGAR	1333	334	20.0%	
DIAB_EYE_EXAM	1333	334	20.0%	
DIAB_NEPHRO_SCREEN	1333	334	20.0%	
HTN_PCP_FOLLOWUP	1334	333	20.0%	
Variant	Mean AUC	Std AUC	Min AUC	Max AUC
Grouped (member_id)	0.9411	0.0030	0.9377	0.9448
Grouped + stratified by care gap	0.9416	0.0024	0.9388	0.9450

5. AUC stability and what we treat as acceptable

For care-gap ranking we treat ~0.70–0.80 as useful for prioritization. Values above 0.90 warrant a time-safety check on reward features.

Variant	Mean AUC	Std AUC	Min AUC	Max AUC
Full features, grouped split, median impute	0.9411	0.0030	0.9377	0.9448
Full features, grouped + stratified	0.9416	0.0024	0.9388	0.9450
No reward features	0.7477	0.0077	0.7372	0.7583
Drop missing prior_closure_rate	0.9273	0.0073	0.9187	0.9386

Per-seed AUC (baseline grouped split)

Seed	AUC	Brier	Precision	Recall
42	0.9448	0.0938	0.8836	0.8337
123	0.9377	0.0963	0.8939	0.8099
456	0.9431	0.0941	0.8737	0.8281

789	0.9414	0.0949	0.8931	0.8153
2026	0.9384	0.0961	0.8751	0.7977

6. Operating threshold vs outreach volume

0.50 is a comparable reporting point, not an operational cut. Choose a threshold from outreach capacity (care managers vs campaigns).

Threshold	Precision	Recall	F1 Score	N Flagged	Pct Flagged
0.3000	0.6798	0.9283	0.7849	1296	54.5%
0.4000	0.8694	0.8419	0.8555	919	38.6%
0.5000	0.8751	0.7977	0.8346	865	36.4%
0.6000	0.9479	0.7292	0.8243	730	30.7%
0.7000	0.9455	0.6944	0.8007	697	29.3%
0.8000	0.9378	0.6038	0.7346	611	25.7%

High / Medium / Low mix on the test set

Tier	N Test	Pct Test	Closure Rate
High	697	29.3%	94.5%
Medium	222	9.3%	63.1%
Low	1459	61.4%	10.3%

Who acts on the score is an InComm operating decision. Short care-manager lists should use a higher threshold (precision). Broader pharmacists or campaign outreach can use a lower threshold (recall). We will lock cut-points once capacity is specified.